

Fixed and Moving Local References in the Rotemberg–HSA Phillips Curve

Restriction Version: F0, U, R1, R2, and R3

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Automatically built from provenance-validated production runs

Abstract

This restriction-version report is separate from `nkpc_hsa_report.pdf`, which remains the historical reduced-form and ablation report. The present document compares a fixed-reference local Rotemberg–HSA benchmark (F0) with an unrestricted moving-reference model (U), the local HSA cross-restricted model (R1), its weak Third-law extension (R2), and a constant-pass-through local benchmark (R3). Every numerical table and figure is generated only from the signed theory-run namespace. Historical artifacts are never substituted for a missing current run.

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1 Research question and evidence ladder

The empirical design separates four logically different objects: fixed-reference theory identities, maintained HSA laws, domain and local-model admissibility, and the moving-reference approximation. The evidence ladder is

$$\text{fixed-reference theory} \rightarrow F0 \rightarrow \text{moving reference} \rightarrow U \rightarrow R1 \rightarrow R2, \quad R3 = R1 + \{\gamma = 0\}.$$

F0 is not nested in U/R1/R2/R3. Within the moving-reference family, $U \supset R1 \supset R2$. R3 is an equality-restricted local benchmark and is not described as a full Co-PaTh model.

2 Current research design: fixed and moving local references

The current specification starts from the quarterly, fixed-reference local Rotemberg–HSA equation. F0 is its fixed- N_0 empirical benchmark. U, R1, R2, and R3 add a slow-moving local reference; this moving-reference step is a semi-structural approximation and is not Eq. 21 itself.

2.1 Restriction taxonomy

Category	Contents
Theory identity / cross-equation relation	$\kappa_{\text{a0_structural}} = (\text{zeta0} - 1) / \chi$; $\Theta_{\text{a0}} = (1 / \chi) * (1 - \rho_{\text{a0}}) / \rho_{\text{a0}}$; $\mu_{\text{a0}} = \text{zeta0} / (\text{zeta0} - 1)$; $d \kappa_{\text{a0}} / d \log N = \text{zeta0} * \Theta_{\text{a0}}$
Maintained HSA law	Second law: $\theta_{\text{a0}} > 0$ and $\kappa_{\text{a0_N_empirical}} > 0$; Weak Third law: $\gamma \leq 0$ (R2 only)
Domain / local admissibility	$\chi > 0$; $\text{zeta0} > 1$; $\mu_{\text{a0}} > 1$; $\kappa_{\text{a0}} > 0$; optional whole-path $\kappa_{\text{a0_t}} > 0$ and $\theta_{\text{a0_t}} > 0$
Moving-reference empirical approximation	moving $N_{\text{star_t}}$ is a slow-moving local reference; $\kappa_{\text{a0_t}} = \kappa_{\text{a0}} + \kappa_{\text{a0_N_empirical}} * N_{\text{bar_t}}$; $\theta_{\text{a0_t}} = \theta_{\text{a0}} + \gamma * N_{\text{bar_t}}$
Deliberately not imposed (future R4)	$\kappa_{\text{a0}} = (\text{zeta0} - 1) / \chi$; $\theta_{\text{a0}} = (10 / \chi) * (1 - \rho_{\text{a0}}) / \rho_{\text{a0}}$

2.2 Model hierarchy and non-nesting

Within the moving-reference family, $U \supset R1 \supset R2$ and R3 is $R1 + \{\gamma = 0\}$. F0 is not nested in that family because its reference is fixed.

Model	Slug	Reference	Cross restriction	Second law	Third law	Gamma	State sampler
F0	hsa_f0	fixed reference	no	maintained	neither	not_applicable	fixed reference ffbs
U	hsa_u	moving reference	no	not maintained	neither	unrestricted	particle gibbs
R1	hsa_r1	moving reference	yes	maintained	neither	unrestricted	particle gibbs
R2	hsa_r2	moving reference	yes	maintained	weak	$\gamma \leq 0$	particle gibbs
R3	hsa_r3	moving reference	yes	maintained	neither	$\gamma = 0$	joint ffbs

2.3 Quarterly structural scaling

With $\bar{N} = 10(\log N - \log N_0)$ and inflation in percentage points, $d\kappa/d \log N = 10\kappa_N$ and the restricted marginal-cost specification has $100\kappa_N = b_x \zeta_0 \theta_0$. The equality is never imposed directly on a gap proxy without an explicit b_x , or on the legacy direct 4Q-YoY regression.

2.4 Current posterior results

CURRENT THEORY ESTIMATES PENDING. Historical tables below are not used as evidence for R1–R3.

2.5 Q/Q versus four-quarter YoY observation design

The two transformations of a common price index are estimated in separate likelihoods and are never stacked as if independent. F0 and U provide the direct observation-design comparison. R1–R3 remain Q/Q-only in this revision because a direct YoY regression does not preserve the meaning of the quarterly structural cross-equation coefficient. **CURRENT OBSERVATION-DESIGN COMPARISON PENDING.**

2.6 Current-run provenance

Status

NO CURRENT RUNS; HISTORICAL ARTIFACTS EXCLUDED

3 Interpretation of nested comparisons

The $U \rightarrow R1$ comparison isolates tension with the local HSA cross-equation restriction. The $R1 \rightarrow R2$ comparison isolates tension with the weak Third law, while $R1 \rightarrow R3$ assesses whether locally constant pass-through is too restrictive. A restriction is not presumed to improve fit. Posterior geometry, convergence, admissibility rejection, and fit must be read together.

4 Observation-design discipline

Quarterly structural frequency is fixed in every run. Q/Q and four-quarter YoY are alternative observation designs and are never stacked in one likelihood. The current direct YoY regression is available only as an F0/U comparison; R1–R3 remain Q/Q-only until a likelihood that aggregates four quarterly structural equations is implemented. Thus the quarterly restriction $100\kappa_N = b_x\zeta_0\theta_0$ is never imposed directly on a reduced-form four-quarter coefficient.

5 Reproducibility and stale-artifact policy

The run manifest records code revision, model definition and hierarchy, exact restrictions, maintained laws, admissibility conditions, inflation and expectation transformations, activity and competition proxies, N_0 , ζ_0/μ_0 treatment, sample, sampler, chains, seeds, and convergence. The report build fails if a comparison would mix incompatible revisions, samples, transformations, observation designs, proxies, priors, or model definitions. The historical report and its result directory are not scanned by this builder.